

DAYUAN CHEN

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[Google Scholar](#) · [GitHub](#) · [Project Page](#)

SUMMARY

Robotics researcher specializing in **embodied AI, robot learning, and force-aware manipulation**, with a focus on combining imitation learning, haptic feedback, and real-time control for long-horizon deformable object manipulation. Full-stack capability in building end-to-end robotic systems spanning teleoperation, policy learning, failure recovery, and deployment on real-world dual-arm platforms. First-author work in **IEEE/ASME Transactions on Mechatronics** and multiple international best-paper awards.

EDUCATION

Ph.D., Robotics — Tohoku University, Japan	10/2022 – 09/2026 (expected)
Dissertation: <i>Force-Aware Dual-Arm System with Learning-Based Control for Garment Manipulation</i>	
M.Eng., Mechanical Engineering — Shenzhen University, China	2016 – 2019
B.Eng., Mechanical Design & Automation — Dalian Minzu University, China	2012 – 2016

SELECTED PUBLICATIONS

First-author

- D. Chen**, K. Tang, Y. Zhang, K. Kosuge, Y. Hirata. "Phase-Conditioned Imitation Learning with Autonomous Failure Recovery for Robust Deformable Object Manipulation." *IEEE/ASME Transactions on Mechatronics*, 2026.
- D. Chen**, Y. Zhang, W. He, et al. "Simplified Offset and Gravity Compensation for Wrist-Mounted F/T Sensors." *IEEE ROBIO* 2025.
- D. Chen**, Z. Liao, et al. "A Human-Guided Approach for Garment Manipulation Using Real-Time Body Tracking and Impedance Control." *IEEE ICARM* 2024.
- D. Chen**, Y. Chen, X. Jiang and Y. Liu, "High-precision end-tip positioning system for automatic interior finishing process based on laser level," *IEEE ROBIO* 2021.
- D. Chen**, K. Li, X. Jiang, Y. Liu. "Development of a Macro-Micro Robot with a Compliant End Effector for Putty Applying." *IEEE RCAR* 2020.

Co-author

- Y. Zhang, **D. Chen**, et al. "An Incremental Hybrid Impedance Algorithm for Stable Force–Position Control." *IEEE TASE* 2025.
- Y. Zhang, **D. Chen**, et al. "Multi-Critic Reinforcement Learning for Garment Handling." *IEEE TASE* 2025.
- X. Huang, **D. Chen**, Y. Guo, X. Jiang, Y. Liu, "Untangling Multiple Deformable Linear Objects in Unknown Quantities With Complex Backgrounds," *IEEE TASE*, 2024
- Z. Liu, **D. Chen**, M. A. Eldosoky, Z. Ye, X. Jiang, Y. Liu, "Puttybot: a sensorized robot for autonomous putty plastering," *Journal of Field Robotics* 2024.

Full list (incl. IROS, CASE, ROBIO, RA-L): [Google Scholar](#)

SELECTED RESEARCH EXPERIENCE

Force-Aware Hierarchical Policy with Failure Recovery — *Doctoral Research* 03/2025 – present

Built a complete force-aware dual-arm manipulation system for long-horizon garment tasks, integrating:

- Phase-conditioned imitation learning policy**: injecting a FiLM layer in ACT encoder for deterministic behavior.
- Visual-force phase estimation**: solving state aliasing problems such as snagging on the hanger.
- Closed-loop failure-recovery mechanism**: automating tasks execution when failure occurs.
- Bilateral haptic-feedback teleoperation system** (Based on Meta Quest 3) with customized vibration feedback.
- Real-time deployment on a DENSO dual-arm platform.
- Raised task success rates from 56%→87% (hanging) and 88%→92% (taking-off)
- Published in *IEEE/ASME Transactions on Mechatronics* (2026).

CMB — Real-Time Cartesian Control Library for ROS 2 — *Open Source* 10/2022 – present

- Developed and **open-sourced C++ easily extensible ROS 2 real-time control architecture** for general-purpose manipulators, with generic kinematics derived from URDF. https://leledeyuan00.github.io/cartesian_motion_base/
- Adopted as the control foundation for several award-winning collaborative studies (see Awards).

Deep / Reinforcement Learning for Deformable Objects

10/2023 – 03/2025

- Deployed an RL policy from a custom PyFlex simulator to the real world (**Sim2Real**) via a human-guided correction approach.
- Built a full dual-arm garment-handling system (ROS 2, Gazebo) with adaptive incremental impedance control for stable force–position interaction.

Macro-Micro Redundant Manipulator for Automated Putty Applying

10/2019 – 01/2022

- Designed the kinematics and control and Gazebo simulation environment for a **9-DOF redundant manipulator** (3-DOF parallel platform + 6-DOF UR arm) for large-area putty applying.
- Developed an **STM32-based embedded subsystem** (single-motor drive, torque sensor, impedance control) integrated into ROS via a custom serial protocol, coordinating low-level force control with high-level motion. (*IEEE RCAR 2020*)
- Contributed to the follow-up system that added mobile-base control and learning-based scratch localization for fully automated operation. (*co-author, Journal of Field Robotics*)

WORK EXPERIENCE

Specially Appointed Researcher — *Industry Collaboration with DAIHEN Corp.*

04/2026 – present

- Applying the force-aware imitation-learning methods from my doctoral research to an **industrial dual-arm cable connector mating task**, collecting demonstration data via bilateral haptic teleoperation and training policies for in-air connector insertion.

Specially Appointed Research Assistant — *TransGP, Hong Kong*

10/2024 – 01/2025

- Built experimental platforms and infrastructure supporting robotic manipulation research, while advancing dual-arm garment manipulation work in collaboration with researchers at the University of Hong Kong.

Research Assistant — *Harbin Institute of Technology*

07/2019 – 08/2022

- Delivered an industrial robotic palletizing system deployed in a real warehouse environment and successfully passed the National Ministry of Science and Technology (MOST) project acceptance. Integrated a vision algorithm (from CUHK) via 3D point clouds from a Smarteyetec camera to detect stacking targets, controlled a SIASUN robot through a custom ROS 2 interface, and connected the system into Cainiao's warehouse pipeline alongside VisionNav's autonomous forklifts.
- Developed ROS projects for KUKA iiwa7 dual-arm robots (custom FRI protocol, Sunrise programming); built lab infrastructure (GitLab, EtherCAT motor drivers).

Embedded Engineer — *Shenzhen MileBot Robotics*

02/2017 – 05/2019

- Developed control algorithms and an FPGA-STM32 (10k Hz versus 1k Hz) embedded system for series elastic actuators in lower-limb rehabilitation exoskeletons (BEAR-H1).
- Contributed to product-level engineering across the full development cycle: **standardized assembly and maintenance documentation**, designed **human-machine interfaces (HMI)** for maintainability, and supported **EMC compliance testing** for a medical-grade device.

SELECTED AWARDS

- **Best Paper in Biomimetics**, IEEE ROBIO 2025 — hybrid robotic hand for fabric manipulation (co-author)
- **Best Conference Paper Finalist**, IEEE ROBIO 2024 — sim-to-real cloth manipulation (co-author)
- **Best Application Video Award**, IEEE CASE 2024 — incremental hybrid impedance control (co-author)
- **Best Conference Paper Finalist**, IEEE ICARM 2024 — human-guided garment manipulation (**first author**)
- **Pioneering Research Support Project for PhD Students** (Tohoku University / JST), 2022–2025 — competitive doctoral fellowship providing full tuition waiver, monthly stipend, and annual research funding

TECHNICAL SKILLS

- **Robot Learning:** Imitation learning (ACT, Diffusion Policy), dataset collection, teleoperation, Sim2Real.
- **Control & Robotics:** Real-time Cartesian / impedance control, ROS/ROS2 (ros_control, MoveIt, Gazebo), dual-arm manipulation, motion planning, sensor processing, kinematics.
- **Programming:** C/C++ (real-time, embedded), Python, PyTorch.
- **Embedded & Hardware:** STM32, FPGA, protocols (EtherCAT, CANopen, Modbus, SPI, I²C, CAN).
- **Tools:** Linux, Git, Docker, LeRobot.

PATENTS

- 4 Chinese patents (invention & utility) in robotic manipulation and end-effector design (co-inventor).